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Started	Sunday, 18 October 2026, 10:44 AM
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Duration	10 mins 6 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It simplifies the sorting process by using only one pass
- c. It avoids the need for recursion
- d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **2**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are split again into smaller sublists
- b. They are combined to form a single sorted list
- c. They are compared element by element
- d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **3**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes data harder to manage
- b. Makes code execution slower
- c. Provides a basis for other algorithms to work efficiently
- d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **4**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Combining sorted sublists
b. Dividing the input into smaller subproblems
c. Comparing each element with the others
d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **5**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort does not guarantee sorted order
b. Bubble Sort cannot handle duplicate elements
c. Bubble Sort is less efficient for large lists
d. Bubble Sort is difficult to implement

The correct answer is: Bubble Sort is less efficient for large lists

Question **6**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Greedy method
b. Divide and conquer
c. Backtracking
d. Dynamic programming

The correct answer is:
Divide and conquer

Question **7**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It compares adjacent elements and swaps them if necessary
b. It divides the list into two halves, sorts each half, and then merges them
c. It repeatedly finds the minimum element and moves it to the sorted part of the list
d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **8**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data sequentially
b. Sorting data in a single pass
c. Rearranging data without sorting
d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Merge Sort
b. Heap Sort
c. Quick Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **10**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Being the most efficient sorting algorithm
b. Bubbling up the largest element to its correct position with each pass
c. Using the divide-and-conquer approach
d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **11**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Sorting the entire list in one pass
b. Bubbling up the smallest element
c. Bubbling up the largest element to its correct position
d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It only works with integer arrays
b. It requires external libraries
c. It is less efficient than custom sorting algorithms
d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **13**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It recursively sorts subproblems
b. It divides the list into sublists
c. It uses the heap data structure
d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **14**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Merge Sort
b. Bubble Sort
c. Quick Sort
d. Linear Search

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Bubble Sort
b. Insertion Sort
c. Merge Sort
d. Quick Sort

The correct answer is: Merge Sort